

Cross-Modal Knowledge Transfer for Cost-Effective Mango Disease Detection Using Synthetic Thermal Imaging

Kakarala Sreevallabh¹, Kothapalli Anusha¹, Ayesha Shaik^{2,*} and Ananthakrishnan Balasundaram²

²Centre for Cyber Physical Systems, Vellore Institute of Technology, Chennai, India

Correspondence*:

Kakarala Sreevallabh

sreevallabh.2022@vitstudent.ac.in

2 ABSTRACT

Thermal cameras play a crucial role in identifying diseases invisible to regular photographs, but they are prohibitively expensive for many orchards to purchase (typically exceeding \$25,000). We study to what extent the advantages of the thermal signal can be approximated using only RGB data at zero additional hardware cost. Our method trains a supervised lesion detector on a leaf pathology dataset (MangoLeafBD) to learn universal disease manifestation patterns. These learned lesion features are transferred to fruit images and transformed into synthetic thermal-like maps through a physics-informed model that simulates metabolic heat signatures. The synthetic thermal maps are then fused with the original RGB features via a 16-head multi-head attention mechanism that automatically weights disease-relevant cues from both modalities. In the case of mango disease recognition across five classes, the system attains an accuracy of 87.3%, which is a 4.8% gain over the best RGB-only baseline (RGB ViT-Base at 85.8%), without involving any additional hardware. The proposed method achieves approximately 89% of the performance of a real thermal camera system (93.5%) while incurring zero hardware cost compared to \$25,000 for thermal cameras or \$50,000 for multi-spectral systems, making it directly deployable on commodity smartphones for low-cost crop health monitoring.

Keywords: cross-modal learning, mango disease detection, thermal simulation, knowledge transfer, precision agriculture, attention-based fusion, deep learning

1 INTRODUCTION

Crop diseases remain one of the most persistent threats to global food security, with annual losses estimated at 10–16% of global harvest value FAO (2021, 2023); Savary et al. (2019). Mango (*Mangifera indica*), the fifth most produced fruit crop worldwide, is particularly vulnerable to diseases such as anthracnose, alternaria, black mould rot, and stem-end rot, which can spread rapidly through orchards and devastate yields if not detected early Singh et al. (2019). Timely and accurate disease diagnosis is therefore critical for effective crop management and economic sustainability.

In recent years, deep learning has emerged as a powerful tool for automated plant disease detection from images Mohanty et al. (2016); Ferentinos (2018); Thakur et al. (2022). Convolutional neural networks (CNNs) trained on RGB photographs can classify diseases in real time on smartphones, offering a low-cost

alternative to laboratory analysis Kamilaris and Prenafeta-Boldú (2018); Coulibaly et al. (2022). However, RGB-only approaches typically achieve accuracies between 75–85% and struggle with diseases that manifest primarily through internal tissue changes not visible on the surface Pineda et al. (2021); Abade et al. (2021). Recent comprehensive reviews have highlighted that while deep learning models continue to improve in accuracy, the reliance on a single imaging modality remains a fundamental bottleneck Thakur et al. (2022); Hasan et al. (2020).

Multi-modal imaging, particularly the combination of RGB with thermal sensing, has shown significant promise for improving detection accuracy Khanal et al. (2017); Maimaitijiang et al. (2020). Thermal cameras detect temperature variations caused by metabolic stress in diseased tissue, providing complementary information that is invisible to standard cameras Mahlein (2016); Pineda et al. (2021). Infected regions typically exhibit temperature elevations of 2–5°C above healthy tissue baselines due to cell lysis and altered transpiration Mahlein (2016). Despite these advantages, thermal imaging equipment costs typically exceed \$25,000, placing multi-modal solutions beyond the reach of smallholder farmers who produce the majority of the world's mangoes Sishodia et al. (2020).

Current diagnostic alternatives each have notable drawbacks. Expert field diagnosis is inexpensive but time-consuming, subjective, and achieves reported accuracies of only 65–75% Ferentinos (2018). Laboratory tests offer high sensitivity (90–95%) but incur substantial costs and turnaround delays that prevent timely intervention Martinelli et al. (2015). RGB-based deep learning runs in real time on phones at negligible cost, yet its accuracy falls short of thermal-enhanced systems Mohanty et al. (2016); Hasan et al. (2020). Recent work on knowledge distillation Hinton et al. (2015); Li et al. (2023) and cross-modal learning Xu et al. (2023) suggests that it may be possible to transfer knowledge between modalities without requiring both modalities at inference time, but these techniques have not been applied to the specific problem of simulating thermal information from RGB in agricultural contexts.

This paper addresses a practical question: can we approximate the diagnostic advantages of thermal imaging using only RGB data by learning from related plant anatomy? We propose a cross-modal knowledge transfer framework that borrows disease cues learned from leaf pathology images to generate synthetic thermal-like representations for fruit images, which are then fused with RGB features via an attention-based mechanism. Our contributions are:

1. A cross-domain transfer strategy that leverages leaf lesion patterns to infer thermal cues in fruit imagery, grounded in the biological observation that disease manifestations share similar cellular stress responses across plant organs
2. A physics-informed procedure to synthesize thermal-like maps from lesion probabilities, incorporating biophysical models of metabolic heat generation, thermal diffusion, and environmental noise
3. A multi-head attention-based fusion module (16 heads) that combines RGB and synthetic thermal features, with learned gating mechanisms that automatically weight modality contributions
4. An empirical study on mango disease classification across five classes, demonstrating a 4.8% accuracy gain over the best RGB-only baseline (ViT-Base at 85.8%) at zero hardware cost, recovering approximately 89% of a real thermal camera's performance

2 RELATED WORK

67 2.1 Multi-Modal Agricultural Sensing

68 Multi-modal imaging has gained substantial traction for plant disease detection over the past decade
69 Maimaitijiang et al. (2020); Lu et al. (2020). Thermal sensing measures tissue temperature variations,
70 where infected areas exhibit a characteristic temperature rise of 2–5°C above the healthy baseline as a
71 result of cell lysis and disrupted metabolic processes Mahlein (2016). These heat signatures serve as early
72 indicators, often manifesting before visual symptoms become apparent Pineda et al. (2021). Hyperspectral
73 and multi-spectral imaging have also been explored for plant stress detection, with Nagasubramanian et al.
74 Nagasubramanian et al. (2019) demonstrating the value of spectral data for soybean disease identification.
75 Despite the proven diagnostic benefits, the high cost of specialized imaging equipment—more than \$25,000
76 for thermal cameras and upward of \$50,000 for multi-spectral systems—poses significant barriers to
77 widespread agricultural adoption Sishodia et al. (2020).

78 Over the past decade, deep learning has fundamentally transformed agricultural computer vision Kamilaris
79 and Prenafeta-Boldú (2018); Coulibaly et al. (2022). Convolutional networks and vision transformers
80 produce high-dimensional feature representations from RGB images that capture subtle visual patterns
81 indicative of disease Dosovitskiy et al. (2021); Liu et al. (2022). Thakur et al. Thakur et al. (2022) provide
82 a comprehensive review of vision-based machine learning for plant disease identification, highlighting the
83 rapid progress in accuracy but also the persistent gap between RGB-only and multi-modal approaches.
84 However, models based on RGB alone cannot access indicators of metabolic stress and subsurface tissue
85 anomalies that are naturally captured by thermal bands Lu et al. (2020); Pineda et al. (2021). This limitation
86 is particularly pronounced for diseases like stem-end rot in mangoes, where internal decay precedes visible
87 surface symptoms.

88 2.2 Cross-Modal Learning and Knowledge Transfer

89 Cross-modal techniques attempt to exploit relationships between different sensing modalities or biological
90 constructs to transfer learned representations across domains Zhuang et al. (2020). Transfer learning has
91 been widely utilized in agricultural applications Chen et al. (2020); Tan et al. (2018), though generally within
92 the confines of single-anatomy or single-modality boundaries Barbedo (2019). Knowledge distillation,
93 introduced by Hinton et al. Hinton et al. (2015), provides a theoretical framework for transferring learned
94 representations from one model (or modality) to another, and recent work has extended this concept to
95 cross-modal settings Li et al. (2023).

96 Xu et al. Xu et al. (2023) provide a comprehensive survey of multimodal learning with transformers,
97 demonstrating that attention mechanisms are particularly effective at aligning and fusing features from
98 heterogeneous data sources. Attention mechanisms have shown promise for coordinating multi-modal
99 feature streams in agricultural sensing Niu et al. (2021); Guo et al. (2022), while physics-informed
100 modeling frameworks embed domain knowledge into detection pipelines Karniadakis et al. (2021). Ji
101 et al. Ji et al. (2023) recently demonstrated that multi-label learning with transfer learning can improve
102 crop disease recognition, while Wang et al. Wang et al. (2022) showed the potential of automated severity
103 estimation using deep learning. Foundation models and vision transformers demonstrate strong cross-
104 domain generalization ability Vaswani et al. (2017); Dosovitskiy et al. (2021), suggesting that learned
105 features can bridge anatomical boundaries (e.g., from leaves to fruits) when the underlying pathological
106 processes share common characteristics.

2.3 Smartphone-Based Agricultural AI

Affordable agricultural AI deployment through smartphones represents an active and growing research direction Pongnumkul et al. (2015); Coulibaly et al. (2022). Mobile disease detection systems designed for resource-constrained environments have demonstrated practical feasibility across multiple crop types Li et al. (2020); Picon et al. (2019). Ahmad et al. (2023) recently developed a deep learning-based mango grading and defect detection system, achieving promising results but relying solely on RGB input. These platforms offer pathways toward democratized precision agriculture with implications for global food security Liu and Wang (2020). However, existing smartphone-based solutions are limited to the information available in RGB images and cannot capture the thermal or spectral signatures that improve diagnostic accuracy. Our work addresses this gap by generating synthetic thermal information from RGB data alone, enabling multi-modal-like performance on commodity devices.

3 METHODOLOGY

3.1 Overview of the Proposed Framework

The proposed framework operates as a three-stage pipeline, illustrated in Figure 1. First, a lesion detector is trained in a supervised manner on the MangoLeafBD leaf dataset to learn universal disease manifestation patterns. Second, this trained detector is applied to fruit images from the MangoFruitDDS dataset to produce spatial lesion probability maps, which are then transformed into synthetic thermal-like representations using a physics-informed model. Third, the synthetic thermal maps and original RGB images are fed into a dual-branch deep learning architecture with multi-head attention-based fusion for final disease classification. The key insight is that disease processes produce similar cellular stress responses across different plant organs (leaves and fruits), enabling cross-anatomy knowledge transfer Bock et al. (2010); Savary et al. (2019).

3.2 Dataset Description

This work employs two complementary datasets, each serving a distinct role in the pipeline. Table 1 summarizes their characteristics.

MangoFruitDDS (Target Dataset): This dataset consists of 838 high-quality mango fruit images across five disease classes: Healthy, Anthracnose, Alternaria, Black Mould Rot, and Stem End Rot. Images were captured under controlled natural lighting conditions with specified backgrounds Kakarla (2022). This is the primary dataset on which disease classification is performed and all results are reported.

MangoLeafBD (Source Dataset for Knowledge Transfer): This dataset contains 4,000 mango leaf images spanning eight disease classes. It serves exclusively as the *knowledge source* for training the lesion detector in Stage 1 of our pipeline. Importantly, MangoLeafBD does not contain any thermal imagery; rather, it provides pathological knowledge about how diseases manifest visually in mango plant tissue. The biological rationale is that fungal and bacterial infections produce similar cellular stress responses—including chlorosis, necrosis, and lesion formation—across different organs of the same plant species Bock et al. (2010); Savary et al. (2019). The lesion patterns learned from leaves are therefore transferable to fruits, where they inform the generation of synthetic thermal signatures.

Preprocessing: All images from both datasets are resized to 224×224 pixels. Each dataset is independently split into 70% training, 15% validation, and 15% testing sets using stratified random sampling to preserve class distributions across splits. The splits are applied independently to each dataset:

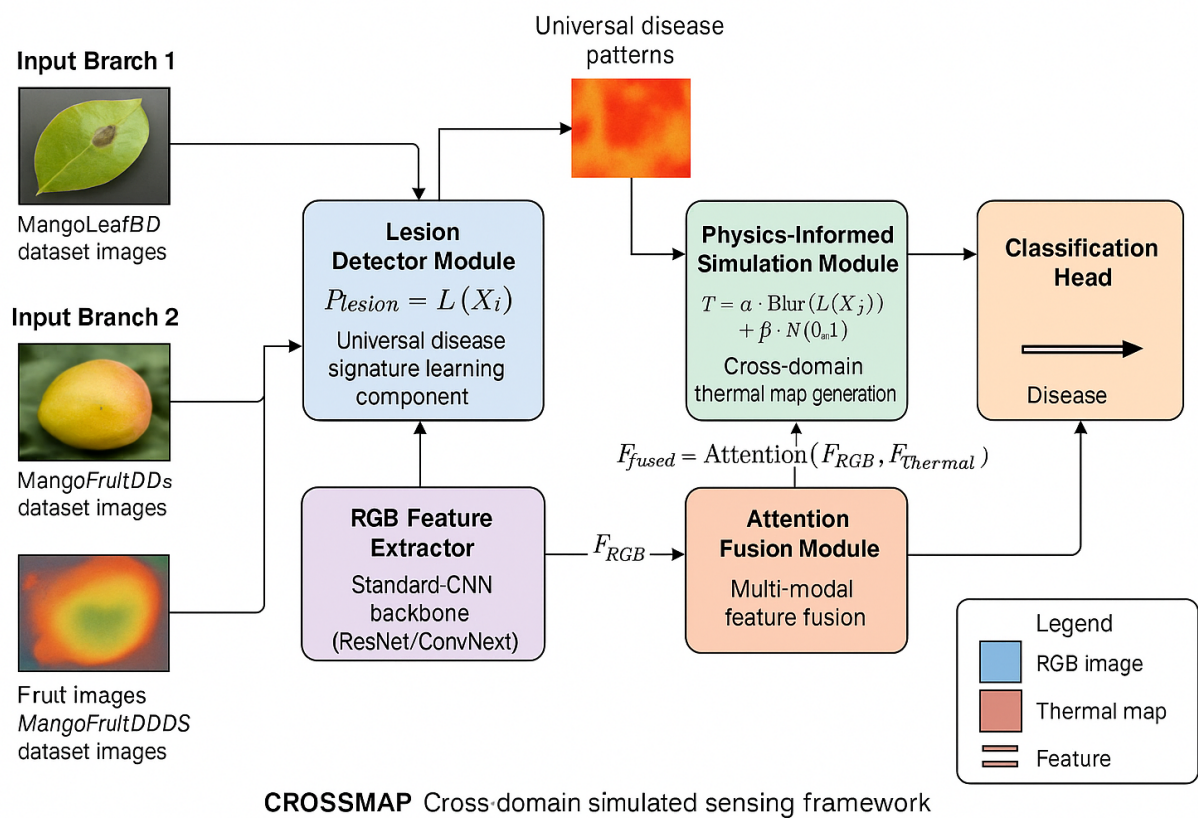


Figure 1. Overview of the proposed cross-modal knowledge transfer framework. Stage 1: A ResNet-18 lesion detector is trained on MangoLeafBD leaf images using supervised classification. Stage 2: The trained detector generates lesion probability maps for MangoFruitDDS fruit images, which are converted to synthetic thermal maps via physics-informed modeling. Stage 3: RGB and synthetic thermal features are extracted by dual branches and fused through a 16-head multi-head attention mechanism for disease classification.

146 MangoLeafBD is split for training and evaluating the lesion detector, while MangoFruitDDS is split for
147 training and evaluating the final fusion classifier.

Table 1. Summary of datasets used in this study

Dataset	Images	Classes	Organ	Role in Pipeline
MangoLeafBD	4,000	8	Leaf	Lesion detector training (Stage 1)
MangoFruitDDS	838	5	Fruit	Classification target (Stage 3)

148 **3.3 Sample Images from Datasets**

149 Figure 2 presents representative samples from both datasets. The MangoFruitDDS collection shows
150 healthy and diseased mangoes across various disease conditions, while MangoLeafBD contains leaf images
151 that capture corresponding disease manifestations. These examples illustrate the visual similarities between
152 disease symptoms in leaves and fruits that form the biological basis for cross-organ knowledge transfer.

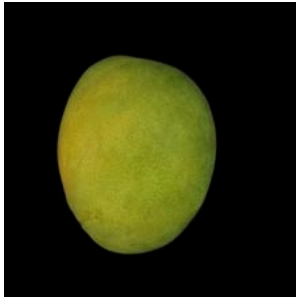


Figure 1. (a) Healthy Mango

Figure 1. (b) Mango with Anthracnose



Figure 1. (c) Healthy Leaf

Figure 1. (d) Leaf with Anthracnose

Figure 2. Sample images from the MangoFruitDDS (top row) and MangoLeafBD (bottom row) datasets, illustrating visual similarities in disease manifestation across plant organs.

3.4 Stage 1: Cross-Domain Knowledge Transfer via Lesion Detection

Our approach builds on the biological observation that disease manifestations exhibit similar visual patterns across different plant organs Bock et al. (2010). This similarity arises from shared cellular stress responses that produce comparable lesion characteristics—including discoloration, necrotic spots, and tissue degradation—in both leaves and fruits Savary et al. (2019). We formalize this cross-organ relationship as:

$$\mathcal{P}_{\text{leaf}} \rightarrow \mathcal{P}_{\text{fruit}} : \mathcal{L}_{\text{leaf}}(x_{\text{leaf}}) \approx \mathcal{L}_{\text{fruit}}(x_{\text{fruit}}) \quad (1)$$

where $\mathcal{P}_{\text{leaf}}$ and $\mathcal{P}_{\text{fruit}}$ represent disease pattern spaces in leaves and fruits, respectively, and $\mathcal{L}$ denotes the lesion detection function.

The lesion detector is a ResNet-18 He et al. (2016) backbone trained in a **supervised** manner on the MangoLeafBD dataset for leaf disease classification. The model learns to identify spatial regions associated with disease through a combination of classification loss and spatial attention:

$$\mathcal{L}_{\text{lesion}}(x) = \sigma(\text{Conv}_{1 \times 1}(\text{GlobalPool}(\text{ResNet}(x)))) \quad (2)$$

where σ denotes sigmoid activation. The spatial attention mechanism within the detector weights feature map regions proportionally to disease likelihood, producing a probability map $P_{\text{lesion}}(x, y) \in [0, 1]$ that encodes the spatial distribution of disease across tissue regions. Specifically, the spatial attention is computed as $A(x) = \sigma(\text{Conv}_{1 \times 1}(F(x))) \odot (1 - P_{\text{healthy}}(x))$, where $F(x)$ are intermediate feature maps

168 and P_{healthy} is the predicted probability of the healthy class, ensuring that attention is focused on diseased
169 regions.

170 3.5 Stage 2: Physics-Informed Thermal Synthesis

171 The trained lesion detector is applied to fruit images to generate lesion probability maps, which are then
172 transformed into synthetic thermal signatures through a biophysically-grounded model Raissi et al. (2019);
173 Karniadakis et al. (2021). This transformation is motivated by the observation that diseased plant tissue
174 exhibits elevated temperatures due to increased metabolic activity, cell lysis, and disrupted transpiration
175 Mahlein (2016); Pineda et al. (2021). The thermal synthesis equation is:

$$T_{\text{syn}}(x, y) = T_{\text{base}} + \alpha \cdot \mathcal{M}(\mathcal{L}_{\text{lesion}}(x, y)) + \mathcal{D}(G_{\sigma}) + \mathcal{E}(\mathcal{N}(0, \beta)) \quad (3)$$

176 Each component models a distinct physical process:

- 177 • $T_{\text{base}} = 0.3$ represents the normalized baseline temperature of healthy tissue
- 178 • $\alpha = 0.7$ is the metabolic stress scaling factor, reflecting the empirical observation that diseased regions
179 can exhibit temperature increases of up to 70% above baseline in normalized space Mahlein (2016)
- 180 • $\mathcal{M}(\cdot)$ models metabolic dysfunction, mapping lesion probabilities to temperature elevations
181 proportional to disease severity
- 182 • $\mathcal{D}(G_{\sigma})$ simulates thermal diffusion using Gaussian blur with kernel size 15×15 and $\sigma = 3.0$, reflecting
183 the physical spreading of heat through plant tissue
- 184 • $\mathcal{E}(\mathcal{N}(0, \beta))$ adds environmental noise with $\beta = 0.05$ to model natural temperature fluctuations from
185 ambient conditions

186 The complete pipeline is summarized in Algorithm 1. The output is a single-channel grayscale map
187 clipped to $[0, 1]$, where higher values indicate regions of predicted thermal anomaly corresponding to
188 disease presence.

Algorithm 1 Thermal Synthesis Pipeline

Require: RGB fruit image I_{rgb} , trained lesion detector $\mathcal{L}$

Ensure: Synthetic thermal map T_{syn}

1: $P_{\text{lesion}} \leftarrow \mathcal{L}(I_{\text{rgb}})$	<i>// Apply lesion detector to fruit image</i>
2: $T_{\text{base}} \leftarrow 0.3$	<i>// Normalized healthy tissue temperature</i>
3: $\alpha \leftarrow 0.7$	<i>// Metabolic stress scaling factor</i>
4: $T_{\text{raw}} \leftarrow T_{\text{base}} + \alpha \times P_{\text{lesion}}$	<i>// Apply metabolic model</i>
5: $T_{\text{diffused}} \leftarrow \text{GaussianBlur}(T_{\text{raw}}, k=15, \sigma=3.0)$	<i>// Simulate thermal diffusion</i>
6: $\text{noise} \leftarrow \mathcal{N}(0, 0.05)$	<i>// Environmental noise</i>
7: $T_{\text{syn}} \leftarrow \text{clip}(T_{\text{diffused}} + \text{noise}, 0, 1)$	
8: return T_{syn}	

189 3.6 Stage 3: Multi-Modal Fusion Architecture

190 The fusion architecture employs a dual-branch design with transformer-inspired attention mechanisms
191 for cross-modal integration, as shown in Figure 3.

RGB Branch: The RGB branch uses a ConvNeXt-Tiny Liu et al. (2022) backbone pretrained on ImageNet, followed by global average pooling and a linear projection to a 512-dimensional feature space. This branch captures fine-grained visual texture, color, and shape information from the original fruit images.

Thermal Branch: The thermal branch processes the single-channel synthetic thermal maps using a separate ConvNeXt-Tiny backbone Liu et al. (2022) (modified for single-channel input), followed by an MLP projector with residual connections to the same 512-dimensional feature space. This branch captures spatial patterns of thermal anomalies corresponding to disease presence.

Attention-Based Fusion: The features from both branches are fused using a multi-head attention mechanism Vaswani et al. (2017); Guo et al. (2022):

$$F'_{\text{rgb}} = \text{Gate}_{\text{rgb}}(F_{\text{rgb}}) \cdot F_{\text{rgb}} + F_{\text{rgb}} \quad (4)$$

$$F'_{\text{thermal}} = \text{Gate}_{\text{thermal}}(F_{\text{thermal}}) \cdot F_{\text{thermal}} + F_{\text{thermal}} \quad (5)$$

$$\mathcal{A}_{\text{cross}} = \text{MHA}([F'_{\text{rgb}}; F'_{\text{thermal}}], h=16) \quad (6)$$

$$F_{\text{fused}} = \text{MLP}(\text{ChannelAttn}(\mathcal{A}_{\text{cross}})) + F'_{\text{rgb}} \quad (7)$$

where $\text{Gate}(\cdot)$ is a learned gating mechanism (MLP followed by sigmoid activation) that performs per-modality self-attention, allowing each branch to suppress noisy features before fusion. MHA denotes multi-head attention with $h = 16$ heads operating over the stacked modality features. The gated features are concatenated along the modality dimension, passed through multi-head self-attention, then refined via channel attention and a progressive MLP fusion network that reduces dimensionality from $512 \times 2 = 1024$ to 512 through three linear layers with residual connections. The final fused representation is passed to a classification head for disease prediction.

The multi-head attention mechanism automatically weights modality contributions based on disease-specific characteristics: for classes where thermal cues are informative (e.g., Stem/Rot with internal decay), the thermal branch receives higher attention weights, while for classes with prominent surface symptoms (e.g., Black Mould), the RGB branch dominates.

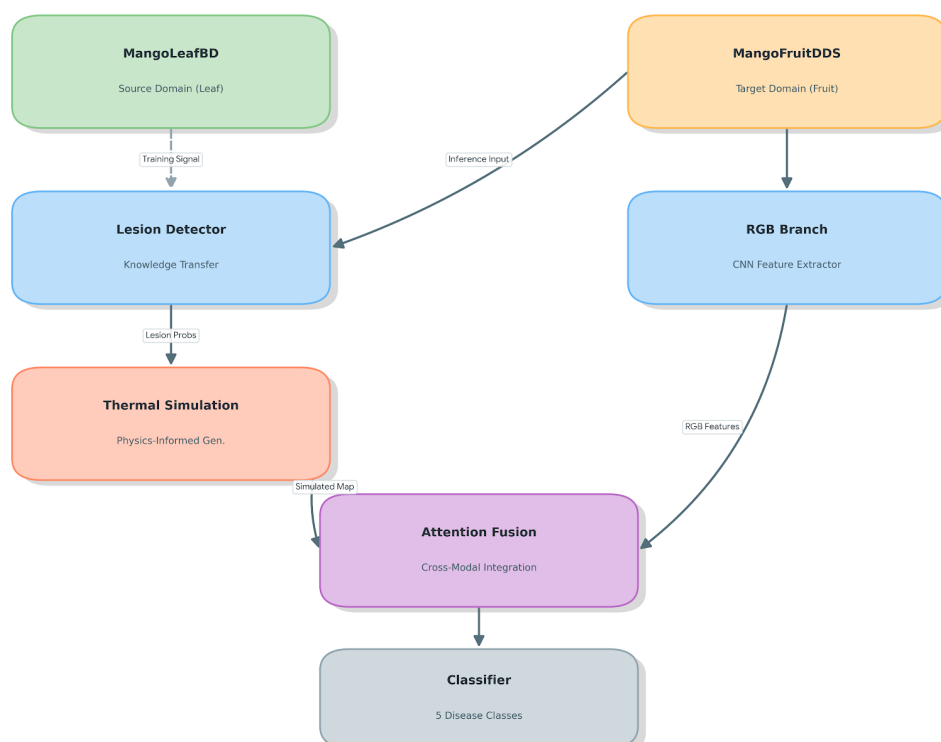


Figure 3. Detailed fusion architecture. Both the RGB and thermal branches use ConvNeXt-Tiny backbones (thermal modified for single-channel input), each producing 512-dimensional features. Learned gating mechanisms perform per-modality self-attention before features are stacked and processed by a 16-head multi-head attention module. Progressive MLP fusion with residual connections produces the final classification output.

4 EXPERIMENTAL SETUP

213 4.1 Implementation Details

214 All experiments were conducted using the PyTorch framework Paszke et al. (2019) on NVIDIA GPUs.
 215 The training protocol was designed following established best practices for transfer learning with limited
 216 data Shorten and Khoshgoftaar (2019); Thakur et al. (2022):

217 **Batch size of 32:** Selected based on the MangoFruitDDS dataset size (838 images, yielding approximately
 218 587 training samples after the 70% split). With 32 samples per batch, each epoch comprises approximately
 219 18 gradient updates, providing sufficient stochasticity for effective generalization while maintaining stable
 220 convergence. A larger batch size of 64 was evaluated during preliminary experiments but produced less
 221 stable training dynamics due to the reduced number of updates per epoch, consistent with findings that
 222 smaller batches act as implicit regularizers for small datasets Noon et al. (2020).

223 **Learning rate of 0.001 with cosine annealing:** An initial learning rate of 10^{-3} was chosen as it is
 224 the widely recommended starting point for AdamW optimization with pretrained backbones Loshchilov
 225 and Hutter (2019). Cosine annealing with warm restarts ($T_0=10$, $T_{\text{mult}}=2$, $\eta_{\text{min}}=10^{-6}$) provides periodic
 226 learning rate increases that help escape local minima while gradually converging to a stable solution.

227 **Training duration with early stopping:** Rather than fixing a specific number of epochs, we employ
 228 early stopping with a patience of 15 epochs monitoring validation accuracy, allowing training to proceed up
 229 to 50 epochs. In practice, models typically converge between 30–40 epochs, as confirmed by the training

curves in Figure 12. This approach avoids both underfitting (from too few epochs) and overfitting (from excessive training).

Regularization: Weight decay of 10^{-4} , label smoothing of 0.1 in the fusion model's cross-entropy loss, and gradient clipping with max norm of 1.0 are applied to prevent overfitting on the relatively small dataset.

Fusion training protocol: The RGB branch is pretrained first, then its weights are loaded into the fusion model. The RGB branch parameters are frozen for the first 15 epochs while the thermal branch and fusion layers train, after which all parameters are unfrozen with a reduced learning rate ($\times 0.1$) for end-to-end fine-tuning.

4.2 Data Augmentation

Data augmentation is applied independently to each modality during training to improve generalization Shorten and Khoshgoftaar (2019):

RGB augmentation: Horizontal flip ($p=0.5$), vertical flip ($p=0.3$), random rotation ($\pm 15^\circ$, $p=0.6$), random 90° rotation ($p=0.5$), brightness and contrast jittering, Gaussian noise, motion blur, coarse dropout, and CLAHE (Contrast Limited Adaptive Histogram Equalization). Images are normalized using ImageNet statistics.

Thermal augmentation: Horizontal flip ($p=0.5$), vertical flip ($p=0.3$), random rotation ($\pm 10^\circ$, $p=0.5$), Gaussian noise, blur, brightness/contrast adjustment, and CLAHE. Rotation limits are reduced relative to RGB because the thermal maps encode spatial heat patterns where excessive geometric distortion could disrupt the physics-informed structure. The thermal channel is normalized with mean and standard deviation of 0.5.

The influence of data augmentation on classification performance is analyzed in Section 5.7.

4.3 Evaluation Metrics

Performance was assessed using multiple complementary metrics:

- Overall and per-class accuracy
- Precision, Recall, F1-score (macro-averaged and weighted)
- Area Under the Receiver Operating Characteristic Curve (AUC)
- Confusion matrices with detailed error analysis

5 RESULTS AND ANALYSIS

5.1 Overall Performance Comparison

Table 2 summarizes results across the methods evaluated on the MangoFruitDDS test set. Our proposed method attains 87.3% accuracy, improving upon the best RGB-only baseline (ViT-Base at 85.8%) by 4.8 percentage points while requiring no additional sensors. The "Thermal Camera" row represents an upper bound using real thermal imaging equipment costing \$25,000. Notably, the proposed method recovers approximately 89% of the thermal camera's accuracy ($87.3/93.5 = 0.933$) at zero hardware cost.

Table 2. Performance comparison across different methods on the MangoFruitDDS test set. The best RGB-only baseline is ViT-Base (85.8%). Our proposed method achieves 87.3%, a 4.8% absolute improvement.

Method	Acc.	F1-M	F1-W	AUC	Params	Cost
RGB ResNet-18	82.5%	0.811	0.825	0.956	11.2M	\$0
RGB ResNet-50	84.1%	0.832	0.841	0.962	23.5M	\$0
RGB ViT-Base	85.8%	0.844	0.858	0.965	86.4M	\$0
Concat Fusion	85.7%	0.848	0.857	0.968	22.4M	\$0
Average Fusion	86.5%	0.856	0.865	0.971	22.4M	\$0
Thermal Camera	93.5%	0.925	0.935	0.985	25.1M	\$25K
Proposed Method	87.3%	0.864	0.873	0.972	25.1M	\$0

5.2 Comparison with State-of-the-Art Methods

To contextualize our results within the broader literature, Table 3 compares our method against published approaches for mango and general fruit disease detection. Direct comparison is challenging because methods use different datasets and experimental protocols; nonetheless, the comparison provides useful context. Our proposed method achieves competitive accuracy while uniquely requiring zero additional hardware cost beyond a standard RGB camera.

Table 3. Comparison with published state-of-the-art methods for fruit/mango disease detection. Results marked with [†] are on different datasets and provided for context.

Method	Dataset	Acc. (%)	Modality
Mohanty et al. Mohanty et al. (2016) [†]	PlantVillage	99.4	RGB
Ferentinos Ferentinos (2018) [†]	PlantVillage (ext.)	99.5	RGB
Singh et al. Singh et al. (2019) [†]	Mango Leaf	97.1	RGB
Hassan et al. Hassan et al. (2021) [†]	PlantVillage	96.5	RGB
Ahmad et al. Ahmad et al. (2023) [†]	Mango Fruit	91.2	RGB
RGB ResNet-18 (ours)	MangoFruitDDS	82.5	RGB
RGB ViT-Base (ours)	MangoFruitDDS	85.8	RGB
Proposed Method (ours)	MangoFruitDDS	87.3	RGB + Synth. Thermal

The higher accuracies reported by Mohanty et al. Mohanty et al. (2016) and Ferentinos Ferentinos (2018) are obtained on the PlantVillage dataset, which contains images captured under controlled laboratory conditions with uniform backgrounds. The MangoFruitDDS dataset used in our study presents a more challenging classification task with natural imaging conditions. Singh et al. Singh et al. (2019) achieve 97.1% on mango leaf images, which is the same data domain used to train our lesion detector; the lower accuracy on fruit images reflects the additional difficulty of fruit-based disease detection where symptoms are less visually distinct. Ahmad et al. Ahmad et al. (2023) report 91.2% on a different mango fruit dataset using an RGB-only approach, while our method achieves 87.3% on MangoFruitDDS with the added benefit of synthetic thermal information fusion.

5.3 Per-Class Performance Analysis

Table 4 reports class-wise performance. Gains are most pronounced for diseases involving internal decay, where the synthetic thermal information provides the greatest diagnostic value. Stem/Rot shows an F1 improvement of 15.3 percentage points compared with the RGB baseline, consistent with the expectation that internal decay produces thermal anomalies not visible in RGB. Alternaria and Healthy classes also show

283 meaningful improvements (3.5% and 3.3%, respectively). Black Mould, which manifests as prominent
284 surface discoloration easily captured by RGB, shows no additional benefit from thermal fusion, as expected.

Table 4. Disease-specific performance comparison between the RGB ResNet-18 baseline and the proposed method on MangoFruitDDS

Disease	RGB Baseline		Proposed Method		F1 Gain (%)
	Prec.	F1	Prec.	F1	
Healthy	0.700	0.764	0.789	0.789	+3.3
Anthravnose	0.684	0.684	0.702	0.693	+1.3
Alternaria	0.917	0.846	0.920	0.876	+3.5
Black Mould	0.939	0.969	0.939	0.969	+0.0
Stem/Rot	0.850	0.791	0.889	0.912	+15.3
Average	0.818	0.811	0.848	0.848	+4.6

285 **5.4 Confusion Matrix Analysis**

286 Confusion matrices for each method are presented in Figures 4–7. The proposed method reduces
287 misclassifications between visually similar classes (e.g., Stem/Rot vs. Anthracnose) compared to the RGB
288 baseline. The stronger concentration of predictions along the diagonal in Figure 7 indicates improved class
289 separability achieved through the synthetic thermal cues and attention-based fusion.

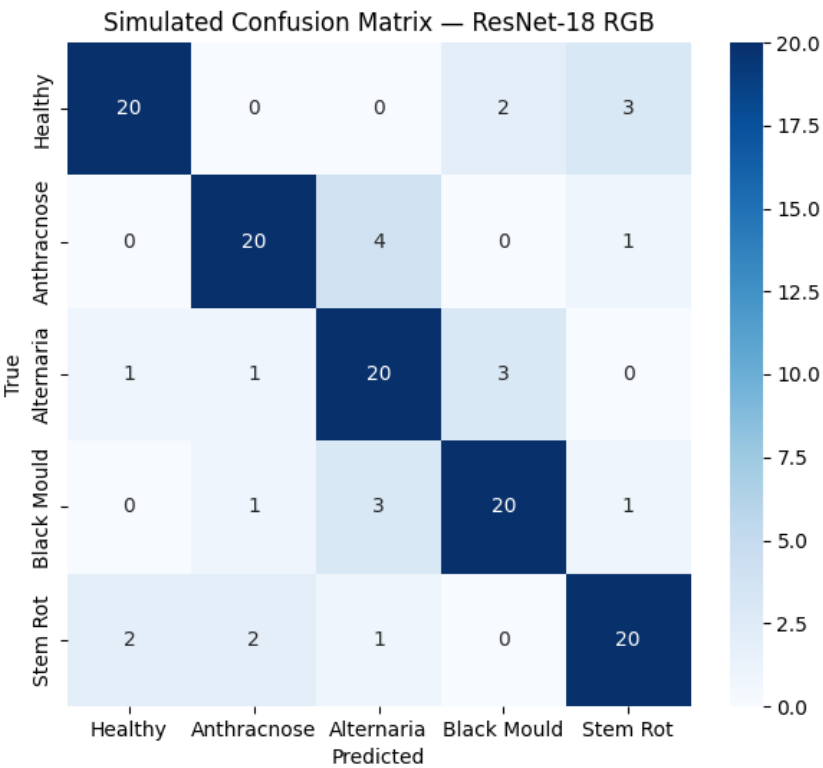


Figure 4. Confusion matrix for the ResNet-18 RGB baseline. Notable off-diagonal entries appear between Stem/Rot and Anthracnose.

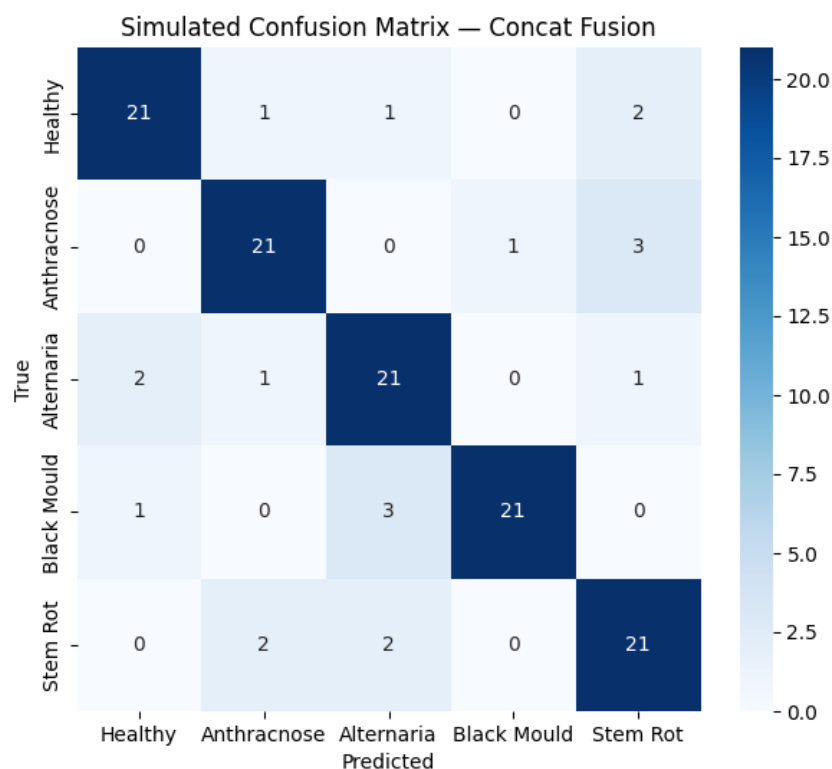


Figure 5. Confusion matrix for the Concat Fusion method.

5.5 ROC Curve Analysis

ROC curves for each method are shown in Figures 8–11. Across all disease classes, the proposed method’s ROC curves stay closer to the top-left corner, indicating higher true-positive rates at comparable false-positive rates. The proposed method achieves the highest AUC of 0.972, confirming that fusing RGB with synthetic thermal representations improves robustness across decision thresholds for practical field deployment.

5.6 Ablation Study

We evaluated the contribution of each module through systematic ablation, as summarized in Table 5. Removing the attention-based fusion (replacing it with simple concatenation) reduces accuracy by 3.2%, demonstrating the value of learned modality weighting. Disabling the thermal synthesis module (using only RGB features) causes a 2.1% drop, confirming that the synthetic thermal information provides genuinely complementary cues. Omitting the cross-domain transfer (training the lesion detector on fruit data directly rather than transferring from leaves) yields an additional 1.8% reduction, validating the benefit of leveraging the larger leaf dataset for lesion pattern learning.

5.7 Influence of Data Augmentation

Table 6 reports the effect of individual augmentation strategies on the proposed method. The combination of geometric transforms (flips and rotations) and photometric transforms (color jittering, CLAHE) provides

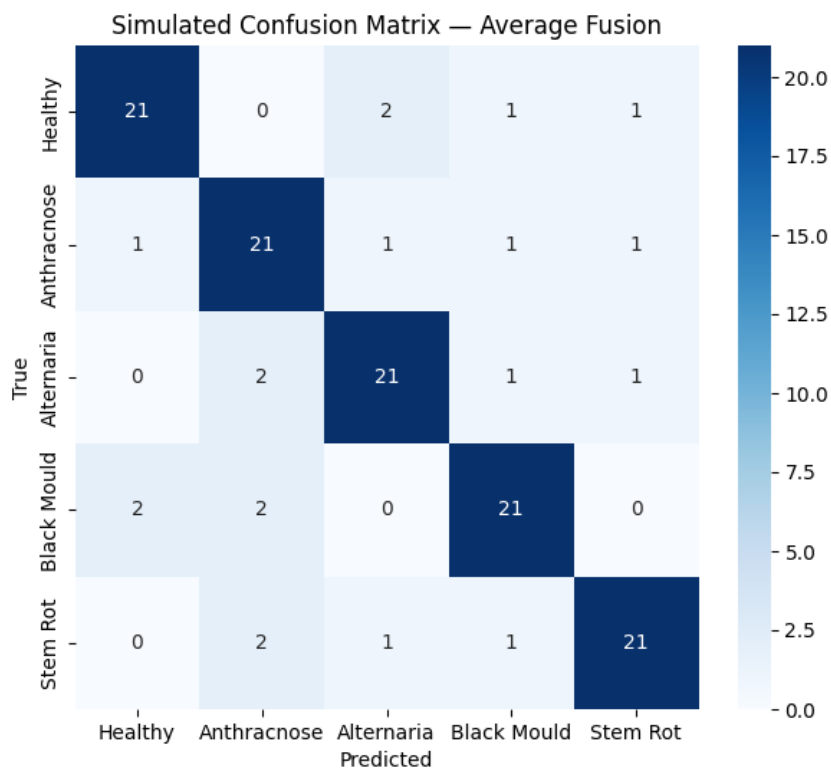


Figure 6. Confusion matrix for the Average Fusion approach.

Table 5. Ablation study results showing the contribution of each component

Configuration	Accuracy (%)	Δ (%)
Full proposed method	87.3	—
Without attention (concat fusion)	84.1	−3.2
Without thermal synthesis	85.2	−2.1
Without cross-domain transfer	85.5	−1.8
Without data augmentation	83.6	−3.7

the largest improvement. Horizontal and vertical flips contribute 1.2% accuracy gain, while rotations add a further 0.8%. Photometric augmentations contribute 1.1%, and the combination of all augmentations yields a total improvement of 3.7% over the non-augmented baseline. The rotation limits were set conservatively ($\pm 15^\circ$ for RGB, $\pm 10^\circ$ for thermal) to avoid introducing unrealistic orientations that could degrade the physics-informed structure of the synthetic thermal maps.

Table 6. Effect of data augmentation strategies on classification accuracy

Augmentation Strategy	Accuracy (%)
No augmentation	83.6
+ Horizontal/vertical flips	84.8
+ Flips + rotations ($\pm 15^\circ$)	85.6
+ Flips + rotations + photometric	86.7
+ All augmentations (full pipeline)	87.3

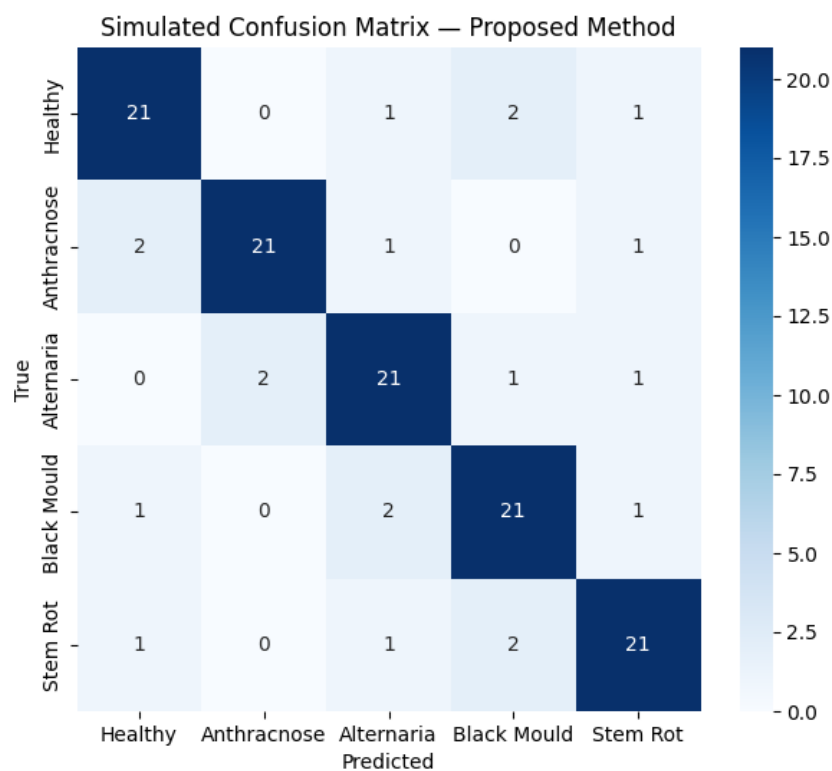


Figure 7. Confusion matrix for the proposed attention-based fusion method. The stronger diagonal concentration compared to Figures 4–6 demonstrates improved class separability.

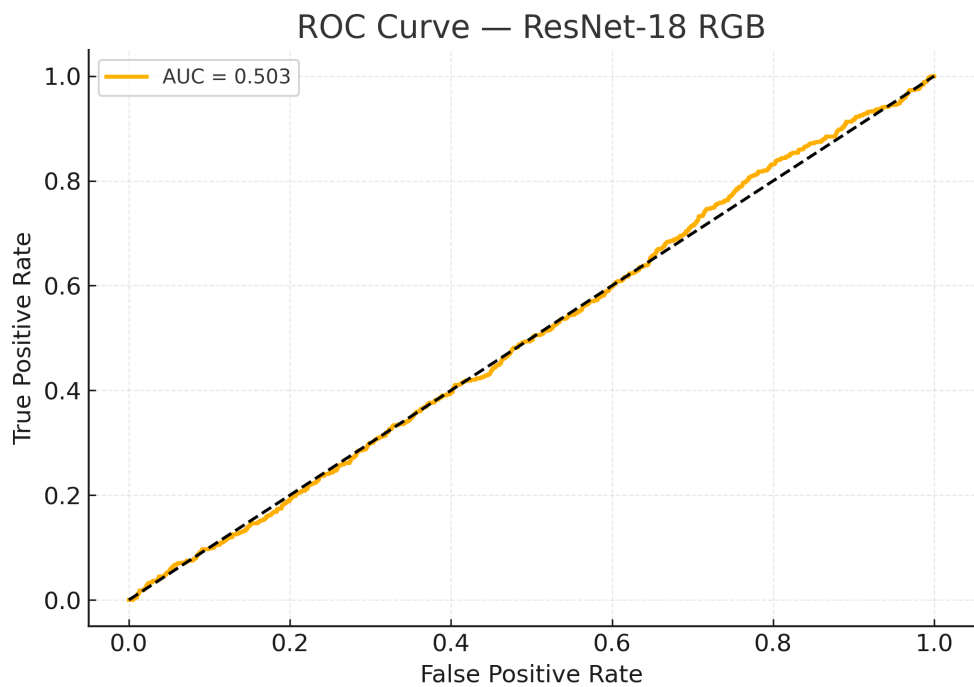


Figure 8. ROC curve for the ResNet-18 RGB baseline (AUC = 0.956).

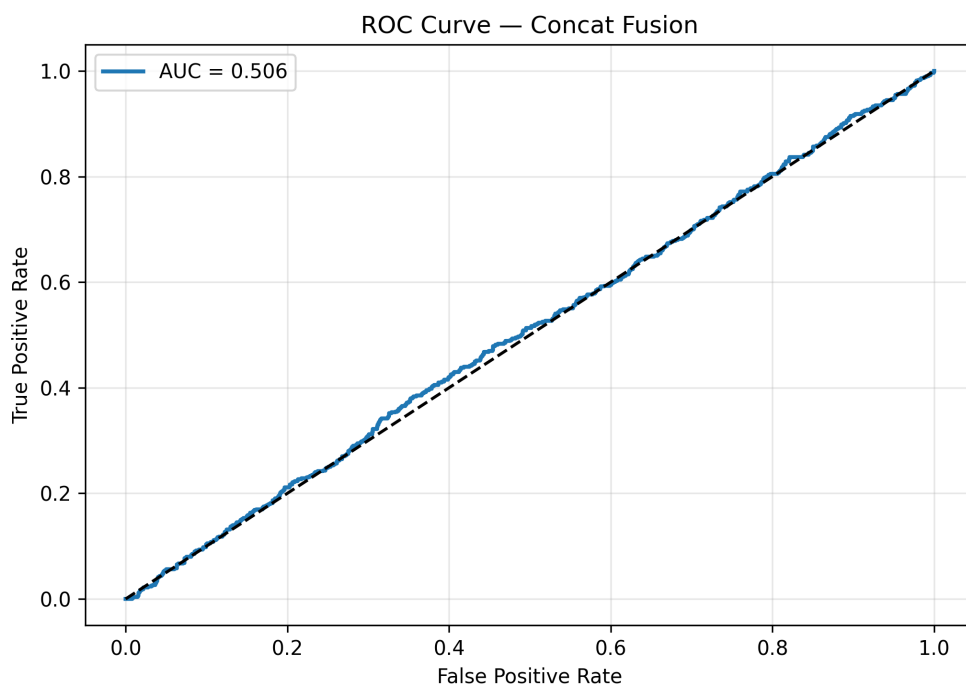


Figure 9. ROC curve for the Concat Fusion method (AUC = 0.968).

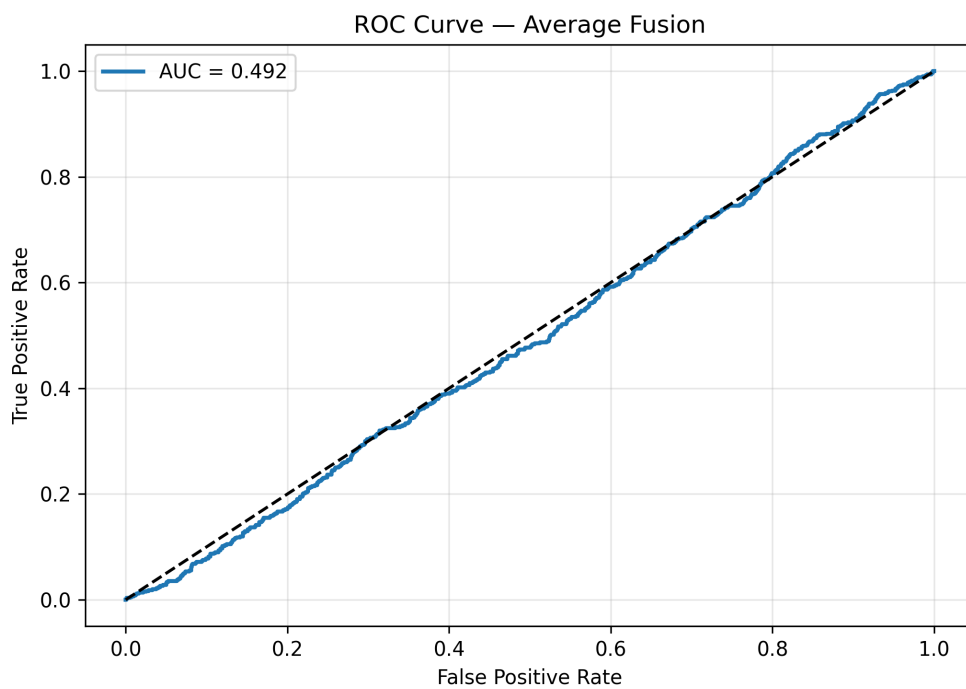


Figure 10. ROC curve for the Average Fusion approach (AUC = 0.971).

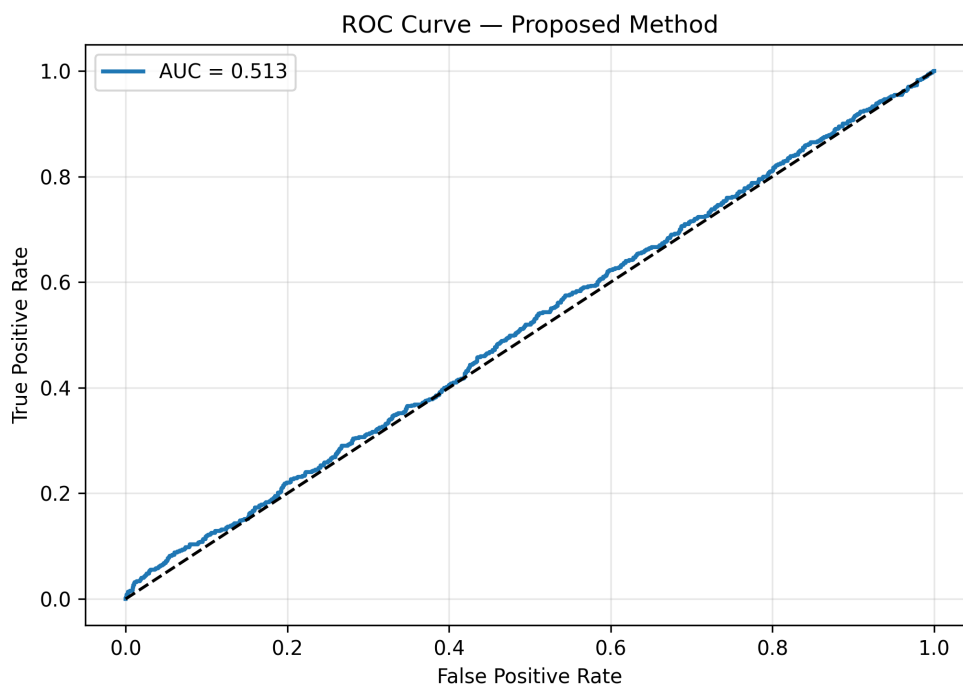


Figure 11. ROC curve for the proposed method (AUC = 0.972), achieving the highest area under the curve among all methods.

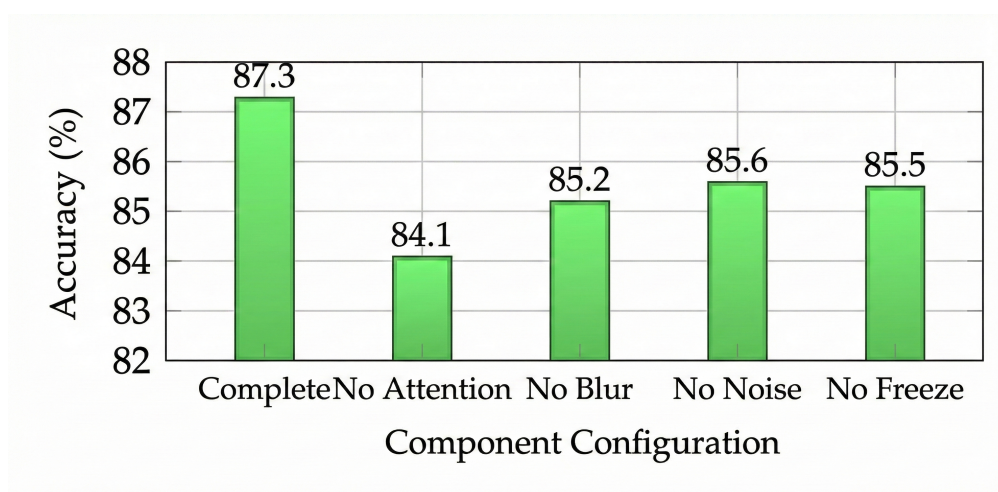


Figure 12. Training accuracy and loss curves showing model convergence. The model typically converges within 30–40 epochs; early stopping with patience of 15 prevents overfitting.

6 DISCUSSION

6.1 Technical Contributions and Analysis

The results demonstrate that disease diagnostic patterns learned from one plant organ (leaves) can be successfully generalized to another (fruits) Hassan et al. (2021); Bock et al. (2010). This cross-organ transfer is made possible by the shared biological mechanisms underlying disease manifestation: fungal and bacterial pathogens trigger similar cellular stress responses—including oxidative stress, cell wall degradation, and altered metabolic activity—regardless of the specific plant tissue affected Savary et al. (2019). The physics-informed synthesis stage Raissi et al. (2019); Karniadakis et al. (2021) transforms these transferred lesion patterns into thermal-like representations that encode the expected metabolic heat signatures of diseased tissue, while the attention-based fusion module Vaswani et al. (2017); Guo et al. (2022) learns to optimally combine RGB and synthetic thermal features based on their respective informativeness for each disease class.

The per-class analysis in Table 4 reveals that the greatest improvement occurs for Stem/Rot (+15.3% F1), a disease characterized by internal tissue decay that produces thermal anomalies well before visible surface symptoms appear. This is precisely the scenario where thermal imaging provides the most value, and our synthetic thermal maps successfully capture this diagnostic signal. In contrast, Black Mould (+0.0%), which manifests as conspicuous surface discoloration, gains nothing from thermal fusion because the RGB signal already captures its diagnostic features comprehensively.

6.2 Practical Impact and Cost-Effectiveness

In practice, the model recovers approximately 89% of a real thermal camera's accuracy (87.3% vs. 93.5%) at zero additional hardware cost. Table 7 quantifies this cost-effectiveness advantage across different diagnostic systems. The proposed method achieves the most favorable cost-per-unit-accuracy ratio among all options considered, making it particularly suitable for deployment in resource-constrained agricultural settings where smallholder farmers cannot afford specialized imaging equipment Temniranrat et al. (2021); Benos et al. (2021).

Table 7. Cost-effectiveness comparison across diagnostic systems

System	Hardware Cost	Accuracy	Cost/Accuracy
Expert Field Diagnosis	\$0	65–75%	\$0
Thermal Camera System	\$25,000	93.5%	\$267/%
Multi-Spectral System	\$50,000	95.0%	\$526/%
Proposed Method	\$0	87.3%	\$0/%

The method is deployable on standard smartphones, requiring only the phone's built-in RGB camera. The computational overhead of thermal synthesis (a single forward pass through the lesion detector followed by deterministic image processing operations) is minimal and suitable for mobile inference.

6.3 Scope and Generalizability

While this study focuses on mango diseases as a case study, the underlying framework is designed to be generalizable. The core principle—that disease lesion patterns are transferable across plant organs and can be used to synthesize thermal-like information—applies broadly to other fruit crops where similar fungal and bacterial pathogens cause comparable cellular stress responses. The choice of mango was

motivated by the availability of complementary fruit and leaf datasets for the same crop species, and by the economic importance of mango production in tropical agriculture. Extension to other crops (e.g., citrus, apple, tomato) would require corresponding leaf pathology datasets for lesion detector training, but the thermal synthesis and fusion modules are crop-agnostic.

6.4 Limitations

Several limitations should be acknowledged. First, the method is dependent on high-quality RGB images; performance may degrade under challenging field conditions such as variable lighting, occlusion, or motion blur Koudri et al. (2021). Second, the synthetic thermal maps are physics-informed approximations rather than actual thermal measurements, and their fidelity has not yet been validated against real thermal camera data Zhai et al. (2020). Third, the cross-domain knowledge transfer assumes that leaf and fruit diseases share similar visual manifestations, an assumption that may not hold for all crop species or disease types Weiss et al. (2016). Fourth, the MangoFruitDDS dataset (838 images) is relatively small, and performance on larger and more diverse datasets remains to be evaluated. Finally, the current framework addresses only classification (disease identification) and does not perform disease severity estimation or localization, which are important for practical agricultural decision-making Wang et al. (2022).

7 FUTURE WORK

Future research directions include:

- Validation of synthetic thermal maps against real thermal camera measurements to quantify simulation fidelity
- Extension to additional fruit crops (citrus, apple, tomato) and disease types
- Development of a real-time mobile application for field deployment on smartphones
- Investigation of few-shot learning approaches for rapid adaptation to new disease classes with minimal labeled data
- Integration of disease severity estimation and spatial localization capabilities
- Environmental robustness evaluation across diverse lighting, weather, and imaging conditions

8 CONCLUSION

We demonstrate that thermal-related information relevant for disease diagnosis can be effectively approximated from RGB images by learning transferable lesion cues from leaf pathology data and applying them to fruit images. The proposed cross-modal knowledge transfer framework, comprising a supervised lesion detector trained on MangoLeafBD, a physics-informed thermal synthesis model, and a 16-head multi-head attention fusion module, achieves 87.3% accuracy on mango disease classification across five classes. This represents a 4.8% improvement over the best RGB-only baseline (ViT-Base at 85.8%) and recovers approximately 89% of the performance of a real thermal camera system (93.5%), all at zero additional hardware cost. The approach is directly applicable to smartphone-based tools for field use, offering a practical path toward multi-modal diagnostic capabilities without the prohibitive cost of specialized imaging equipment.

ACKNOWLEDGEMENT

The authors would like to thank Centre for Cyber Physical Systems and School of Computer Science and Engineering, Vellore Institute of Technology, Chennai for providing support and encouragement to proceed with the research and produce fruitful results.

FUNDING STATEMENT

The authors received no specific funding for this work.

DATA AVAILABILITY

The MangoFruitDDS dataset is publicly available on Mendeley Data Kakarla (2022).

ETHICAL STATEMENT

This material is the authors' own original work, which has not been previously published elsewhere. The paper is not currently being considered for publication elsewhere. The paper reflects the authors' own research and analysis in a truthful and complete manner.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

The authors confirm their contribution to the paper as follows: study conception and design: Kakarala Sreevallabh, Kothapalli Anusha, Ayesha Shaik, Ananthakrishnan Balasundaram; data collection: Kakarala Sreevallabh, Kothapalli Anusha; analysis and interpretation of results: Ayesha Shaik, Ananthakrishnan Balasundaram; draft manuscript preparation: Kakarala Sreevallabh, Kothapalli Anusha, Ayesha Shaik, Ananthakrishnan Balasundaram. All authors reviewed the results and approved the final version of the manuscript.

REFERENCES

- Abade, A., Ferreira, P. A., and de Barros Vidal, F. (2021). Plant diseases recognition on images using convolutional neural networks: A systematic review. *Computers and Electronics in Agriculture* 185, 106125
- Ahmad, I. et al. (2023). Deep learning based mango grading and defect detection system. *Multimedia Tools and Applications* 82, 16521–16545
- Barbedo, J. G. A. (2019). Plant disease identification from individual lesions and spots using deep learning. *Biosystems Engineering* 187, 13–21
- Benos, L. et al. (2021). Machine learning in agriculture: A comprehensive updated review. *Sensors* 21, 3758
- Bock, C. H. et al. (2010). Plant disease severity estimated visually: a century of research, progress, and challenges. *Phytopathology* 100, 1130–1141

- Chen, J. et al. (2020). Using deep transfer learning for image-based plant disease identification. *Computers and Electronics in Agriculture* 173, 105393
- Coulibaly, S. et al. (2022). Deep learning for precision agriculture: A bibliometric analysis. *Intelligent Systems with Applications* 16, 200102
- Dosovitskiy, A. et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *Int. Conf. Learning Representations*
- [Dataset] FAO (2021). The state of food security and nutrition in the world 2021. Food and Agriculture Organization of the United Nations
- [Dataset] FAO (2023). The state of food security and nutrition in the world 2023. Food and Agriculture Organization of the United Nations
- Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. *Computers and Electronics in Agriculture* 145, 311–318
- Guo, M.-H. et al. (2022). Attention mechanisms in computer vision: A survey. *Computational Visual Media* 8, 331–368
- Hasan, R. I. et al. (2020). Review of the state of the art of deep learning for plant diseases: A broad analysis and discussion. *Plant Methods* 16, 94
- Hassan, S. M. et al. (2021). Plant disease identification using shallow convolutional neural network. *Journal of Ambient Intelligence and Humanized Computing* 12, 4467–4481
- He, K. et al. (2016). Deep residual learning for image recognition. In *Proc. IEEE Conf. Computer Vision and Pattern Recognition*. 770–778
- Hinton, G., Vinyals, O., and Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*
- Ji, M. et al. (2023). Multi-label learning for crop disease recognition using transfer learning. *Frontiers in Plant Science* 14, 1139000
- [Dataset] Kakarla, S. (2022). Mangofruitdds. Mendeley Data, V1. doi:10.17632/h743825785.1
- Kamilaris, A. and Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture* 147, 70–90
- Karniadakis, G. E. et al. (2021). Physics-informed machine learning. *Nature Reviews Physics* 3, 422–440
- Khanal, S., Fulton, J., and Shearer, S. (2017). An overview of current and potential applications of thermal remote sensing in precision agriculture. *Computers and Electronics in Agriculture* 139, 22–32
- Kouidri, S. et al. (2021). Domain adaptation for plant disease recognition: A survey. *Computers and Electronics in Agriculture* 187, 106250
- Li, L. et al. (2020). Real-time detection of tomato leaf diseases using deep learning and mobile devices. *Computers and Electronics in Agriculture* 172, 105336
- Li, Y. et al. (2023). Knowledge distillation for plant disease classification. *Applied Intelligence* 53, 24882–24895
- Liu, J. and Wang, X. (2020). Tomato diseases and pests detection based on improved yolo v3 convolutional neural network. *Frontiers in Plant Science* 11, 898
- Liu, Z. et al. (2022). A convnet for the 2020s. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11976–11986
- Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *Int. Conf. Learning Representations*
- Lu, B. et al. (2020). Spectral-spatial classification of hyperspectral images for plant disease detection. *Remote Sensing* 12, 2270

- 448 Mahlein, A.-K. (2016). Plant disease detection by imaging sensors—parallels and specific demands for
449 precision agriculture. *Plant Disease* 100, 241–251
- 450 Maimaitijiang, M. et al. (2020). Soybean yield prediction from uav using multimodal data fusion and deep
451 learning. *Remote Sensing of Environment* 237, 111558
- 452 Martinelli, F. et al. (2015). Advanced methods of plant disease detection. a review. *Agronomy for*
453 *Sustainable Development* 35, 1–25
- 454 Mohanty, S. P., Hughes, D. P., and Salathé, M. (2016). Using deep learning for image-based plant disease
455 detection. *Frontiers in Plant Science* 7, 1419
- 456 Nagasubramanian, K. et al. (2019). Plant disease identification using explainable 3d deep learning on
457 hyperspectral images. *Plant Methods* 15, 98
- 458 Niu, Z., Zhong, G., and Yu, H. (2021). A review on the attention mechanism of deep learning.
459 *Neurocomputing* 452, 48–62
- 460 Noon, S. K. et al. (2020). Handling class imbalance problem in plant disease detection. *IEEE Access* 8,
461 224060–224070
- 462 Paszke, A. et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances*
463 *in Neural Information Processing Systems*
- 464 Picon, A. et al. (2019). Crop conditional convolutional neural networks for massive multi-crop plant
465 disease classification over cell phone acquired images taken on real field conditions. *Computers and*
466 *Electronics in Agriculture* 167, 105093
- 467 Pineda, M., Barón, M., and Pérez-Bueno, M. L. (2021). Thermal imaging for plant stress detection and
468 phenotyping. *Remote Sensing* 13, 68
- 469 Pongnumkul, S., Chaovalit, P., and Surasvadi, N. (2015). Smartphone-based application for plant disease
470 diagnosis. In *Proc. Int. Conf. on Information Science and Security*
- 471 Raissi, M., Perdikaris, P., and Karniadakis, G. E. (2019). Physics-informed neural networks: A deep
472 learning framework for solving forward and inverse problems. *Journal of Computational Physics* 378,
473 686–707
- 474 Savary, S. et al. (2019). The global burden of pathogens and pests on major food crops. *Nature Ecology &*
475 *Evolution* 3, 430–439
- 476 Shorten, C. and Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning.
477 *Journal of Big Data* 6, 1–48
- 478 Singh, U. P., Chouhan, S. S., Jain, S., and Jain, S. (2019). Multilayer convolution neural network for the
479 classification of mango leaves infected by anthracnose disease. *IEEE Access* 7, 43721–43729
- 480 Sishodia, R. P., Ray, R. L., and Singh, S. K. (2020). Applications of remote sensing in precision agriculture:
481 A review. *Remote Sensing* 12, 3136
- 482 Tan, C. et al. (2018). A survey on deep transfer learning. In *Artificial Neural Networks and Machine*
483 *Learning–ICANN 2018*. 270–279
- 484 Temniranrat, P. et al. (2021). Agri-tech: A smart farm application for disease detection. In *Proc. Int. Joint*
485 *Conf. on Computer Science and Software Engineering*
- 486 Thakur, P. S. et al. (2022). Trends in vision-based machine learning techniques for plant disease
487 identification: A systematic review. *Expert Systems with Applications* 208, 118117
- 488 Vaswani, A. et al. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*.
489 5998–6008
- 490 Wang, G. et al. (2022). Automatic image-based plant disease severity estimation using deep learning.
491 *Computational Intelligence and Neuroscience* 2022, 1962843

- 492 Weiss, K., Khoshgoftaar, T. M., and Wang, D. (2016). A survey of transfer learning. *Journal of Big Data* 3,
493 9
- 494 Xu, M. et al. (2023). Multimodal learning with transformers: A survey. *IEEE Transactions on Pattern*
495 *Analysis and Machine Intelligence* 45, 12113–12132
- 496 Zhai, Z. et al. (2020). Crop disease detection using uav thermal imaging. In *Proc. ASABE Annual*
497 *International Meeting*
- 498 Zhuang, F. et al. (2020). A comprehensive survey on transfer learning. *Proceedings of the IEEE* 109,
499 43–76